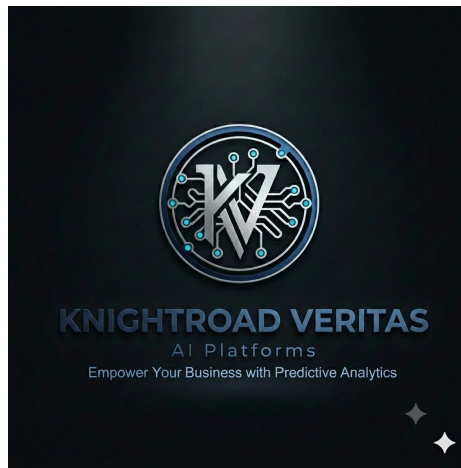




KnightRoad Veritas AI Construction Platform

Design Studio — How-To User Guide

KnightRoad Veritas AI Platforms | July 2026

Document type	End-user how-to guide
Module	Design Studio (/design-studio)
Audience	Project managers, designers, instructors, students
Format	Step-by-step instructions with screen mockups

Introduction

The **Design Studio** lets you create building designs directly inside the Veritas platform — no external CAD software required. Draw floor plans in 2D, preview the building in 3D, name rooms, lay out MEP runs, and export industry-standard IFC4 files or push models to Speckle for team collaboration.

What you can do

- Draw walls, columns, doors, windows, and MEP duct/pipe runs on a snap grid
- Edit any element by dragging — walls, endpoints, columns, openings, MEP paths
- Auto-detect rooms with area calculation; name and type each space
- Build multi-storey designs with independent plans per floor
- Preview in 3D with construction-phase filtering
- Save to project, export IFC4, or push to Speckle

Quick start (5 minutes)

1. Open **Design Studio** from the dashboard navigation.
2. Select your project and load a template (e.g. Vocational / Academic).
3. Draw or edit walls to define rooms; add doors and windows.
4. Click inside a room to name it (e.g. "Workshop A" · Office).
5. Switch to **3D View** to preview the extruded model.
6. Click **Save**, then **Export IFC** to download a BIM file.

Tools reference

Tool	Action	Finish / cancel
Select / Move	Click to select; drag to move walls, columns, openings, MEP	Delete key removes selection
Wall	Click start, click end (chains segments)	Esc stops chaining
Column	Click to place	—
Door / Window	Click on a wall	—
MEP run	Click to add path points	Double-click or Esc
Delete	Click element or opening to remove	—

1. Getting Started — Workspace Overview

The Design Studio is an in-browser floor-plan editor that turns your drawings into 3D BIM geometry. Open it from the top navigation bar at **/design-studio** or click **Design Studio** on the Executive Dashboard.

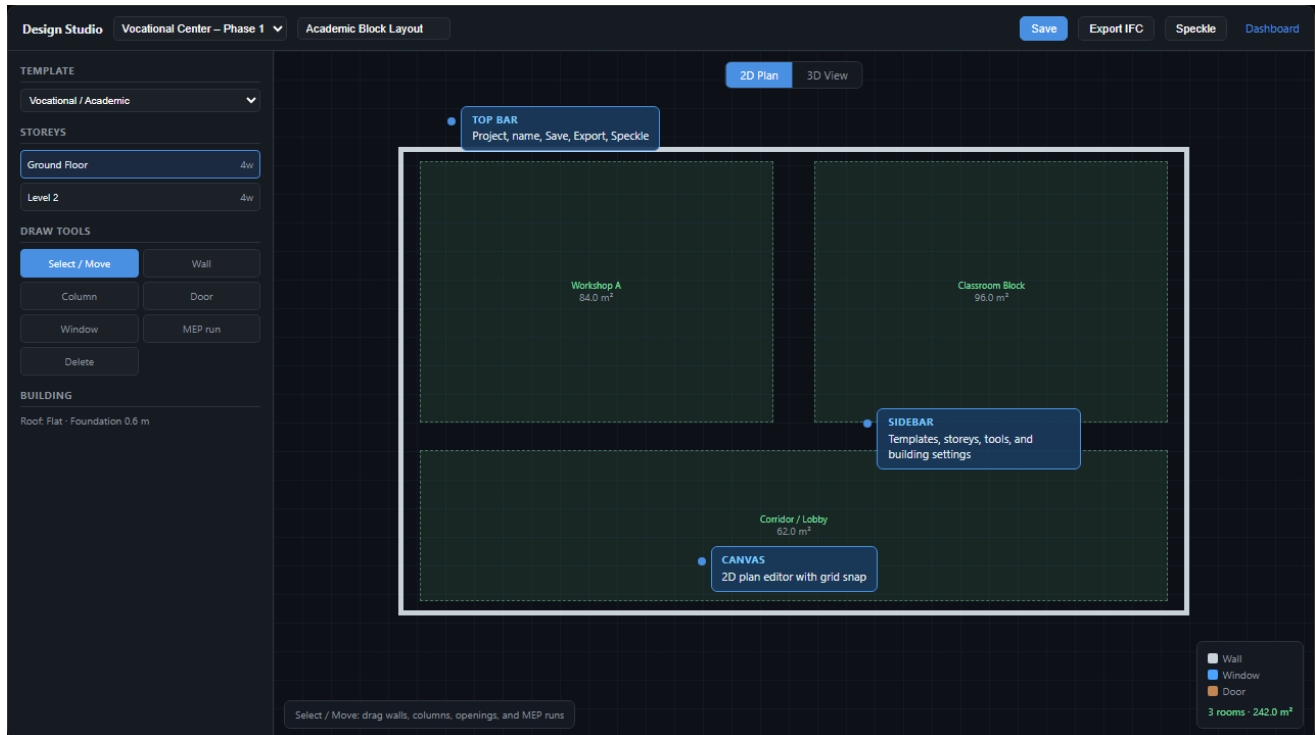


Figure — Getting Started — Workspace Overview

Steps

1. **Select a project** from the dropdown in the top bar. The studio saves designs against this project — choose an active project before saving or exporting.
2. **Name your design** using the text field next to the project selector.
3. **Load a template** (optional) from the sidebar — eight building types seed an editable perimeter-wall plan you can modify freely.
4. Use the **2D Plan** tab to draw and edit; switch to **3D View** at any time for an instant preview.

Tips

- Scroll the mouse wheel to zoom; right-click and drag (or middle-mouse drag) to pan the canvas.
- Enable **Snap to 0.5 m grid** for precise, construction-friendly dimensions.

2. Drawing Walls

Walls are the foundation of every design. Use the **Wall** tool to draw straight segments that form the building footprint and interior partitions.

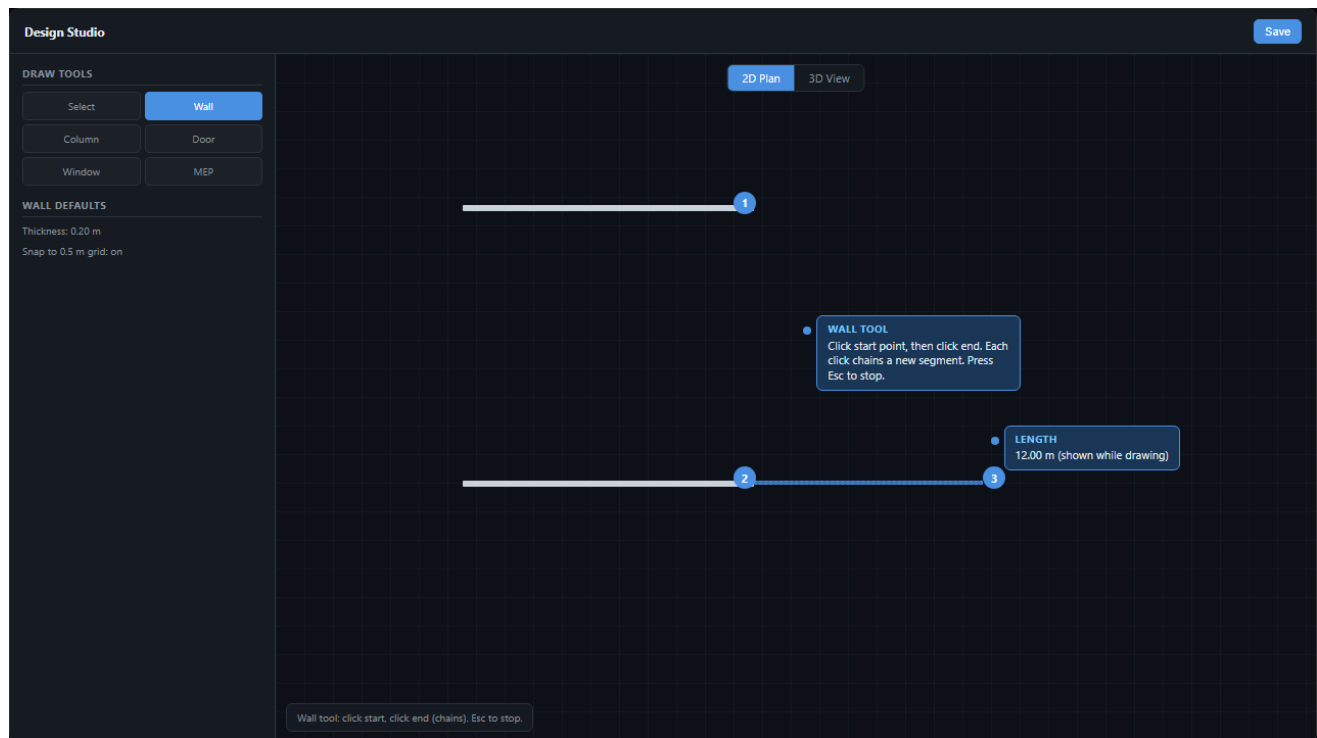


Figure — Drawing Walls

Steps

1. Click the **Wall** tool in the sidebar Draw tools panel.
2. Click on the canvas to set the **start point** of the first wall segment.
3. Click again to set the **end point**. A wall is created and the end point becomes the start of the next segment — you can chain walls continuously.
4. Press **Esc** to stop chaining and exit wall-drawing mode.
5. Adjust default **wall thickness** in the Wall / opening defaults panel before drawing.

Tips

- Close loops of walls to define rooms — the editor auto-detects enclosed spaces.
- Minimum wall length is 0.05 m; very short segments are ignored.

3. Doors & Windows

Place openings directly on walls. Doors and windows are cut into the wall geometry in 3D (sill, header, and glass or door leaf) and exported as typed IFC elements.

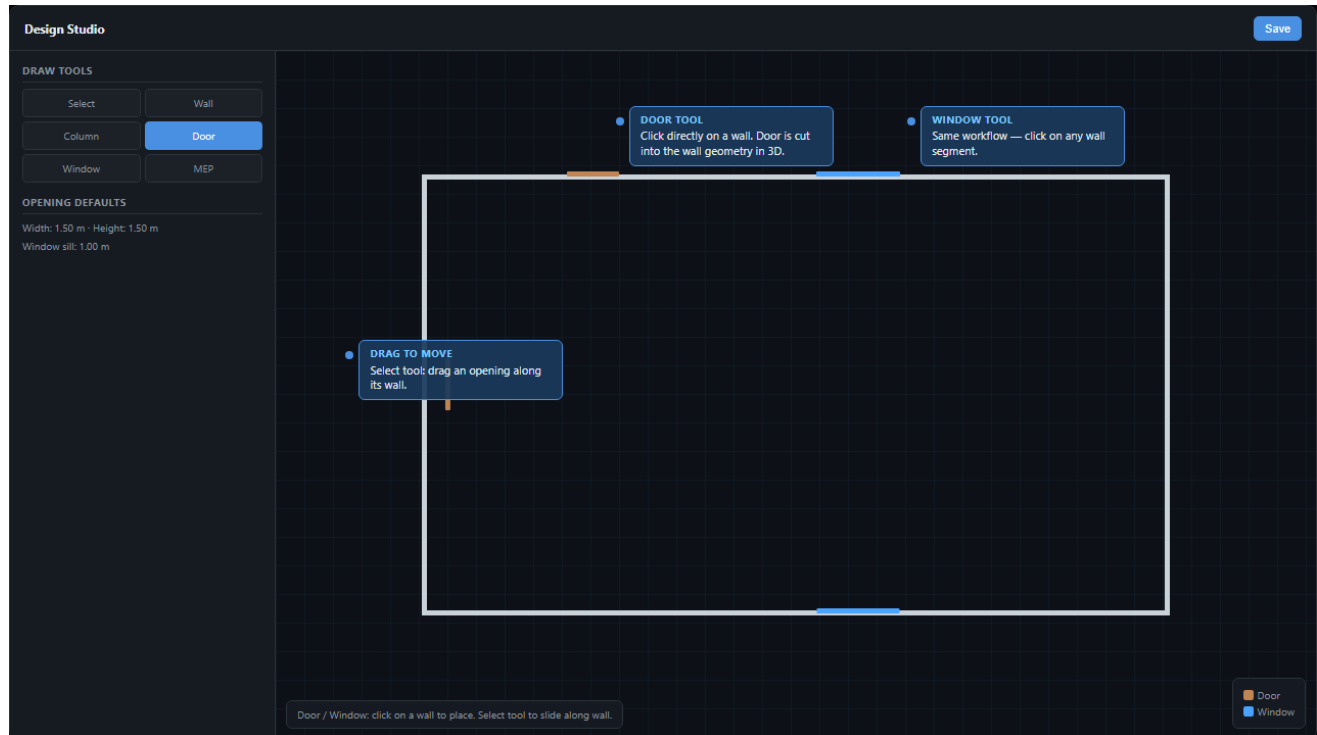


Figure — Doors & Windows

Steps

1. Select the **Door** or **Window** tool.
2. Click on an existing wall segment. The opening is placed at that location.
3. Set default **width**, **height**, and (for windows) **sill height** in the sidebar before placing.
4. Switch to **Select / Move** and drag an opening along its wall to reposition it.
5. Use the **Delete** tool or **Delete** key to remove an opening.

Tips

- Doors default to floor level (sill 0 m); windows default to a 1.0 m sill.
- Ground-floor front doors can be placed on any exterior wall segment.

4. Columns & MEP Runs

Add structural columns at grid points and lay HVAC duct or plumbing pipe runs along multi-point paths. MEP appears in 3D just under the ceiling and exports as IfcDuctSegment / IfcPipeSegment.



Figure — Columns & MEP Runs

Steps

1. **Columns:** Select the Column tool and click once on the plan to place a column.
2. **MEP runs:** Select the MEP run tool. Choose Duct or Pipe and set size and height above floor in the MEP defaults panel.
3. Click to add points along the run path (corners, branches, etc.).
4. Double-click or press **Esc** to finish the run.
5. In Select mode, drag a **vertex** to reshape a run, or drag the **run body** to move the entire path.

Tips

- Default duct size: 0.40 m; default pipe size: 0.15 m.
- MEP height is measured from the floor of the active storey.
- Filter to the MEP phase in 3D View to inspect ducts and pipes in isolation.

5. Select & Move Elements

The Select / Move tool lets you edit existing geometry without redrawing. Drag walls, columns, openings, and MEP runs directly on the plan.

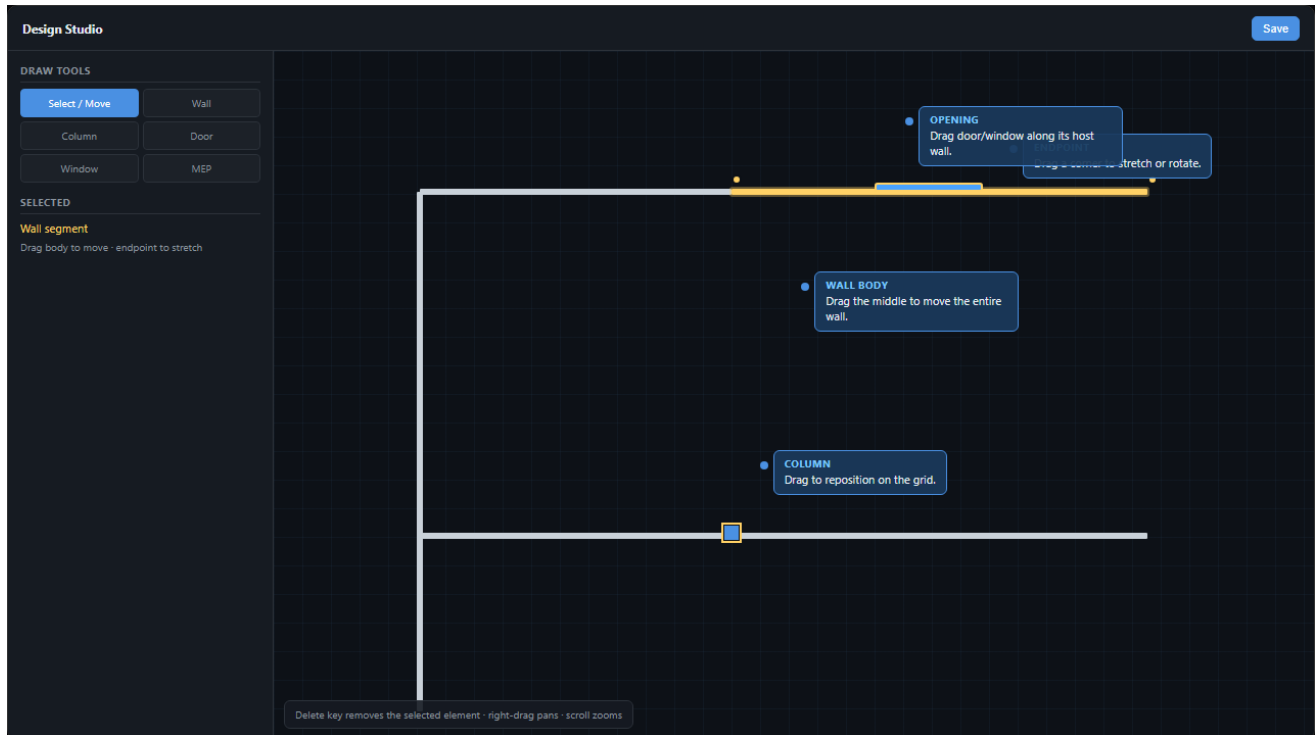


Figure — Select & Move Elements

Steps

1. Click **Select / Move** in the Draw tools panel.
2. **Wall endpoint:** Click near either end of a wall and drag to stretch or rotate.
3. **Wall body:** Click the middle of a wall and drag to move the entire segment.
4. **Column:** Click a column and drag to a new grid position.
5. **Opening:** Click a door or window and drag along its host wall.
6. **MEP vertex:** Click a corner point on a duct/pipe run and drag to reshape.
7. **MEP run:** Click the run body and drag to translate the whole path.
8. Press **Delete** or **Backspace** to remove the selected element.

Tips

- Selected elements highlight in gold (#ffd166).
- Room selection does not use Delete — rooms are derived from wall geometry.

6. Room Auto-Detection & Labelling

When walls form closed loops, the editor automatically detects interior rooms, calculates area, and displays labels on the plan. Name and type each room for IFC export as `IfcSpace`.

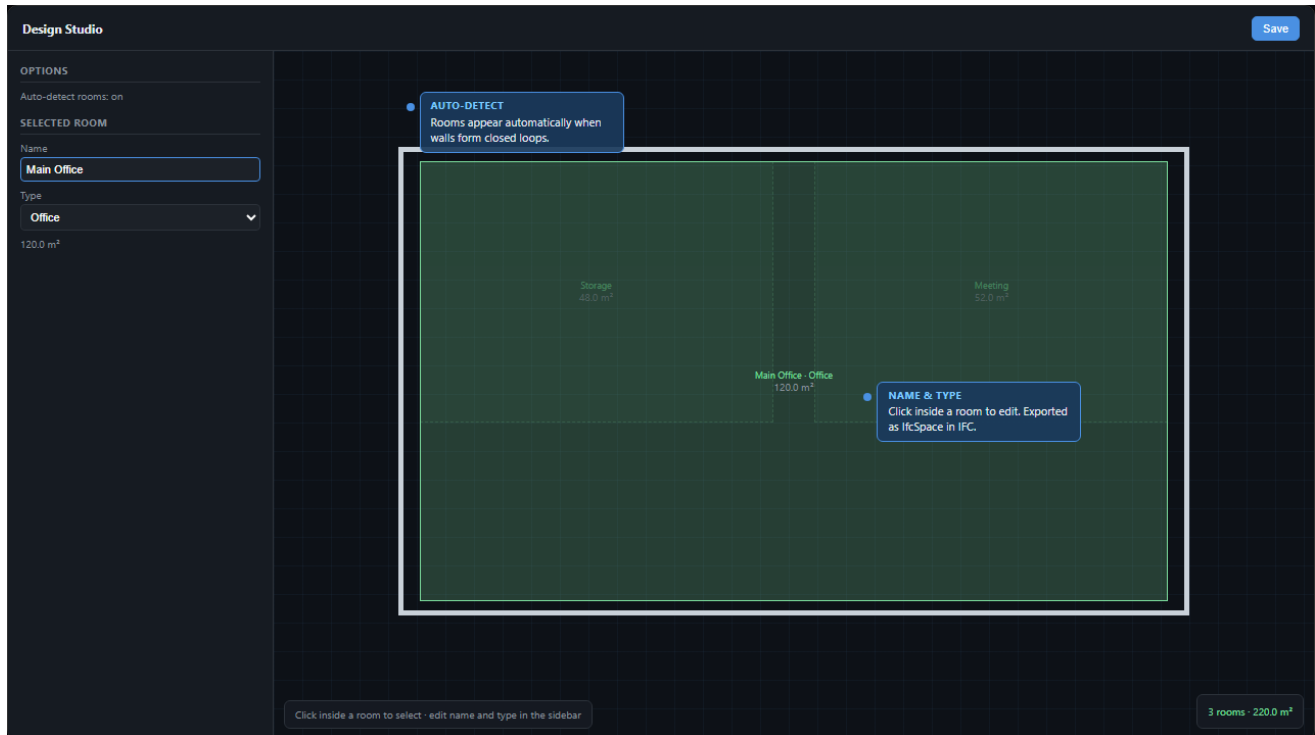


Figure — Room Auto-Detection & Labelling

Steps

1. Ensure **Auto-detect rooms** is checked in the sidebar (on by default).
2. Draw walls that form closed perimeters — rooms appear with a green fill and area label.
3. Click inside a room with the **Select / Move** tool to select it.
4. Edit the **Name** and **Type** (Living, Kitchen, Office, etc.) in the Selected room panel.
5. Room names and types persist across edits and are included when you Save or Export IFC.

Tips

- The legend shows total room count and combined floor area for the active storey.
- Room `ObjectType` in the IFC file matches the Type dropdown (e.g. Office → `IfcSpace.ObjectType`).
- Uncheck Auto-detect rooms to hide room overlays while editing complex geometry.

7. Multi-Storey Editing

Each building can have up to 40 storeys, each with its own independent floor plan, height, slab thickness, walls, columns, MEP, and rooms.

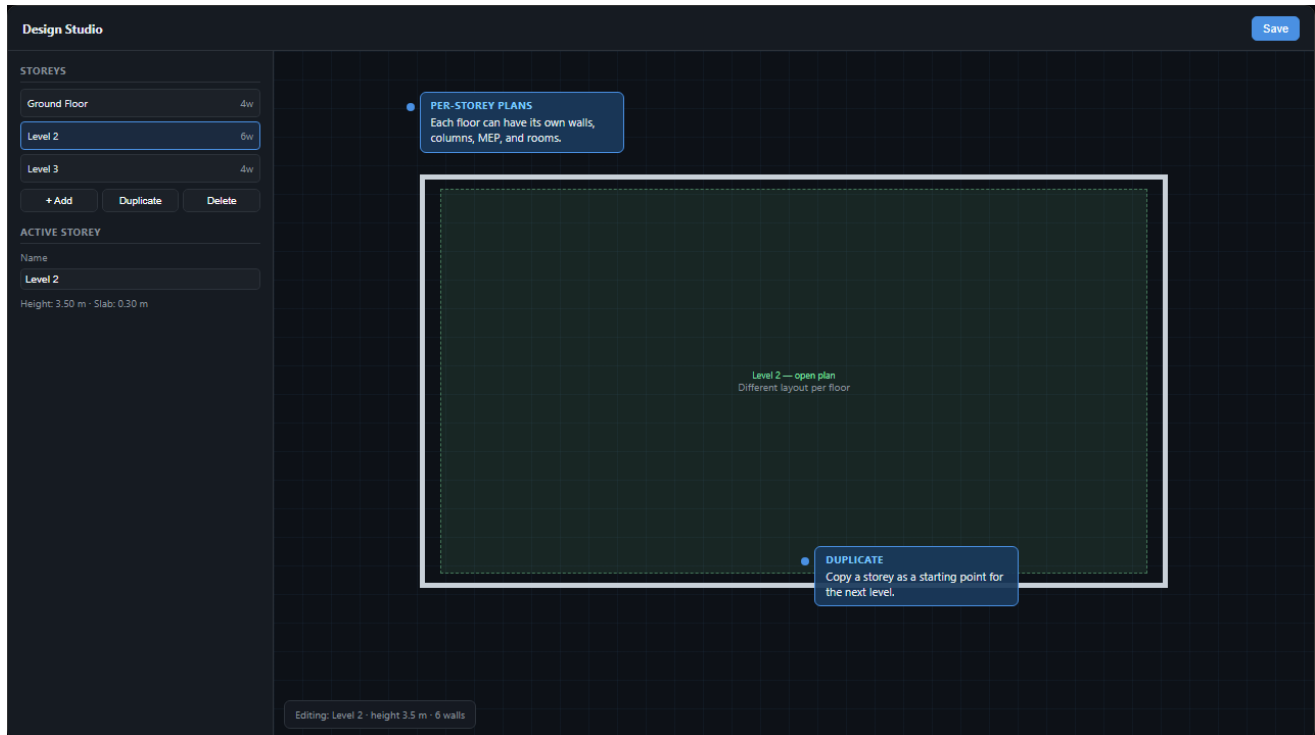


Figure — Multi-Storey Editing

Steps

1. Use the **Storeys** list in the sidebar to switch between floors.
2. Click **+ Add** to create a new empty storey.
3. Click **Duplicate** to copy the active storey's plan (walls, columns, MEP) as a starting point for the next level.
4. Click **Delete** to remove a storey (at least one storey is required).
5. Set the active storey's **name**, **height (m)**, and **slab thickness** in the fields below the storey list.

Tips

- Each storey shows its wall count (e.g. '4w') in the list for quick reference.
- Roof, foundation depth, and margin apply to the whole building, not per storey.

8. 3D Preview & Phase Filtering

Switch to the 3D View tab for an instant extruded preview of your plan. Filter by construction phase to inspect foundation, structure, MEP, cladding, or finishing elements separately.

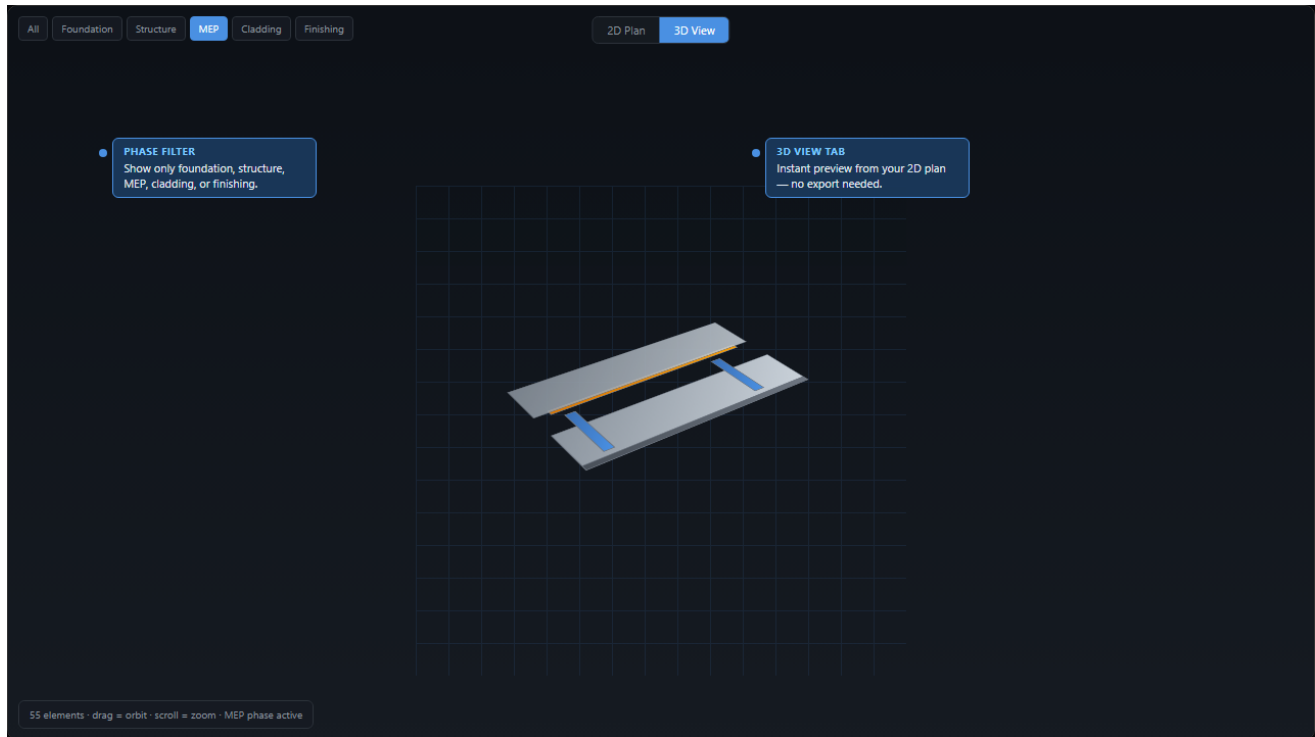


Figure — 3D Preview & Phase Filtering

Steps

1. Click the **3D View** tab at the top of the canvas area.
2. Drag the mouse to **orbit** the camera around the model.
3. Scroll to **zoom** in and out.
4. Use the **phase buttons** (Foundation, Structure, MEP, Cladding, Finishing) to show only elements in that construction stage.
5. Click **All** to restore the full model.

Tips

- The 3D preview updates automatically when you edit the 2D plan.
- Saved designs also appear in the Executive Dashboard 3D viewer when the project has no uploaded IFC/DWG file.

9. Save, Export IFC & Push to Speckle

Three actions connect your design to the rest of the Veritas platform and external BIM tools.

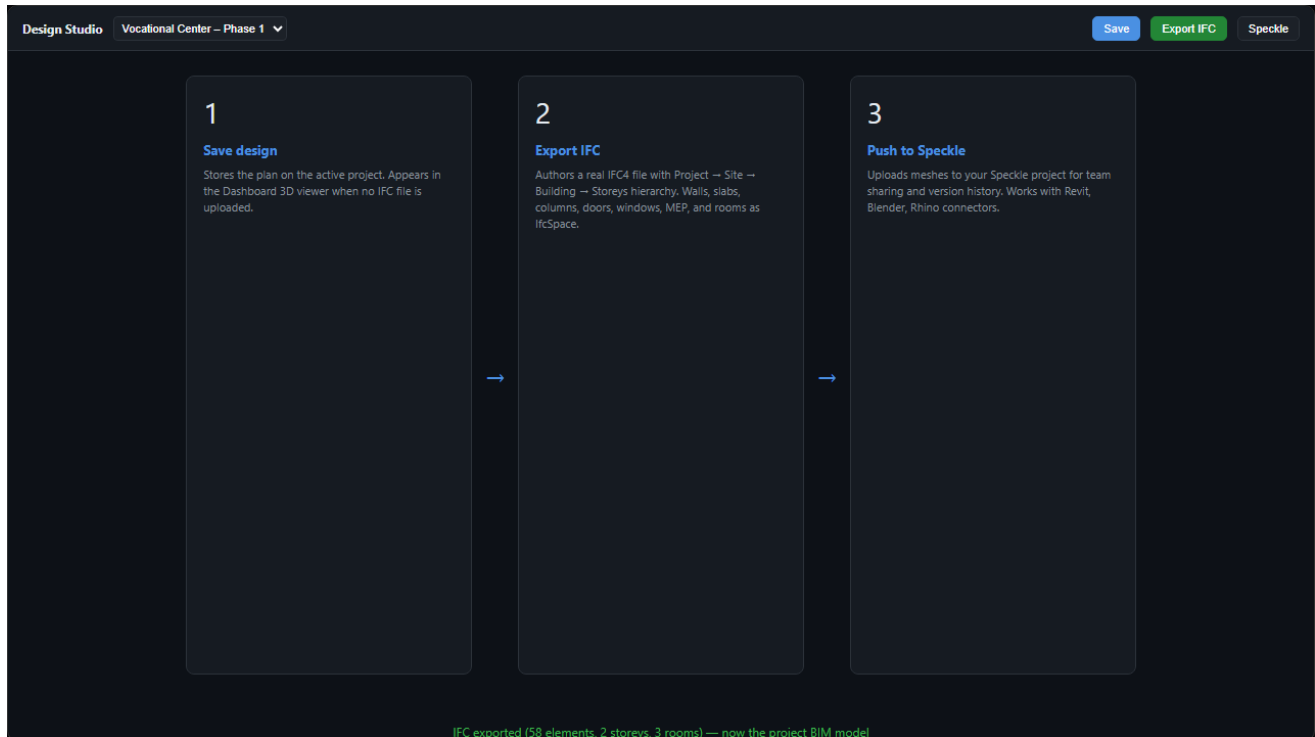


Figure — Save, Export IFC & Push to Speckle

Steps

1. **Save** — Writes the design spec to the active project. Required before IFC export or Speckle push. The dashboard 3D viewer picks up saved designs automatically.
2. **Export IFC** — Authors a real IFC4 file with full spatial hierarchy (Project → Site → Building → Storeys) and typed elements: IfcWall, IfcSlab, IfcColumn, IfcWindow, IfcDoor, IfcDuctSegment, IfcPipeSegment, IfcRoof, IfcFooting, and IfcSpace (rooms). Registers as the project's BIM model and opens the download.
3. **Push to Speckle** — Converts the design to Speckle meshes and creates a version on your Speckle project. Enter a Speckle project (stream) ID and token when prompted, or use the server's configured SPECKLE_TOKEN.

Tips

- IFC files import into Revit, ArchiCAD, Blender (Bonsai), and other IFC-aware tools.
- Speckle enables round-trip collaboration with external authoring tools.

Keyboard & Mouse Reference

Input	2D Plan view	3D View
Left-click	Tool action (draw, place, select)	—
Left-drag	Move selected element (Select tool)	Orbit camera
Right-drag / middle-drag	Pan canvas	—
Scroll wheel	Zoom in / out	Zoom in / out
Esc	Stop wall chain / finish MEP run	—
Delete / Backspace	Remove selected element	—

Workflow integration

Dashboard 3D viewer — After Save, open the Executive Dashboard for the same project. If no IFC/DWG file is uploaded, your saved design renders automatically in the BIM viewer with phase filtering.

IFC pipeline — Export IFC registers the file as the project's BIM model. Re-import via the standard IFC upload path or open in Revit/ArchiCAD/Blender.

Speckle — Push to Speckle creates a versioned cloud model. Link a Speckle project under the project's Speckle settings, then pull geometry via `/api/project/<id>/speckle-geometry`.

Troubleshooting

- **No rooms detected** — Walls must form closed loops. Check for gaps at corners; snap-to-grid helps align endpoints.
- **Save disabled / no project** — Select an active project from the top-bar dropdown.
- **IFC export fails** — Save the design first. Ensure ifcopenshell is installed on the server.
- **Speckle push fails** — Provide a valid Speckle token and project (stream) ID.
- **3D view empty** — Draw at least one wall or load a template to generate geometry.